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SHORT-PAPER

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LLM as User Simulator: Towards Training News Recommender without Real User Interactions

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Abstract

News recommendation systems traditionally rely on extensive real user interaction data to personalize content, which is often inaccessible and raises privacy concerns, particularly in regions lacking such data. To address these challenges, we propose **LAUS** (**L**LM **A**s **U**ser **S**imulator), a novel framework that leverages LLM to simulate user interactions for training a news recommender without real user data. Our framework consists of two stages. First, we simulate user clicks by using an LLM to generate click probabilities for candidate news articles based on simulated user reading histories, with candidate sets generated through random sampling and embedding-based retrieval. Second, we train a lightweight news recommender using these simulated user interactions. Experiments on three datasets demonstrate that a news recommender trained with simulated data outperforms models using LLM prompting, while significantly reducing recommendation time per user. Our findings highlight the potential of using LLMs as user simulators to address data scarcity in news recommendation systems.

CCS Concepts

• **Information systems → Personalization**.

Keywords

news recommendation, large language model, user simulation

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1 Introduction

In the digital age, news recommendation systems are essential for helping users navigate the vast amount of information available online by delivering personalized content that matches their interests [5, 13, 35]. Training these systems typically relies on substantial amounts of real user interaction data, such as click logs, to accurately model user preferences. However, only large-scale platforms like Microsoft Bing have access to such extensive datasets, and collecting this data raises privacy concerns related to the collection and use of personal information. This lack of data poses a significant barrier to developing and refining news recommendation systems, especially for languages and regions where user interaction logs are not readily available. Therefore, research on zero-shot news recommendation techniques or on training news recommenders without real user data is important.

Recently, the remarkable capabilities of Large Language Models (LLMs) [1, 7, 19, 26], which embody extensive world knowledge and can perform a wide array of tasks [24, 31], have led to their widespread adoption in recommendation systems [3, 16–18, 20, 29, 38]. Their ability to understand context and generate relevant content without relying on prior user interaction data makes them well-suited for zero-shot recommendation scenarios, leading to more active research in this area [4, 11, 28, 30, 40]. However, directly employing LLMs for news recommendation poses challenges due to their substantial size, resulting in high inference costs. Using LLM prompting for recommendations requires significant computational resources for each inference, making it difficult to deploy in real-world scenarios where efficiency and scalability are critical.

To address these challenges, we introduce a novel framework named **LAUS** (**L**LM **A**s **U**ser **S**imulator), which utilizes an LLM to simulate user interactions for training a news recommender. In the simulation stage, we iteratively generate user click histories by predicting click probabilities for candidate news articles based on the simulated user's reading history. This involves sampling candidate articles through random selection and embedding-based retrieval, and then using the LLM to estimate click probabilities for these candidates, effectively simulating user choices without the need for real click logs. In the training stage, we train a news recommender using the rich probability distributions obtained from the LLM, rather than relying on binary click indicators. Not only does this two-stage framework eliminate the dependency on large-scale user interaction data, but it also addresses the inefficiencies associated with directly using LLMs at inference time.

To demonstrate the effectiveness of our proposed framework, we conducted experiments on three datasets: MIND [37], Adressa [8], and EB-NeRD [14]. Our news recommender, trained on LAUS-simulated data, achieved high performance despite not utilizing real user interaction data. Specifically, compared to news recommendation using LLaMA-70B prompting, our model achieved an average of 12% higher performance in terms of MRR, while reducing the time required to provide recommendations to a single user by a factor of 625. Additionally, we observed that the model trained with both our LAUS-simulated data and real user interactions outperformed a model trained solely on real user interactions.

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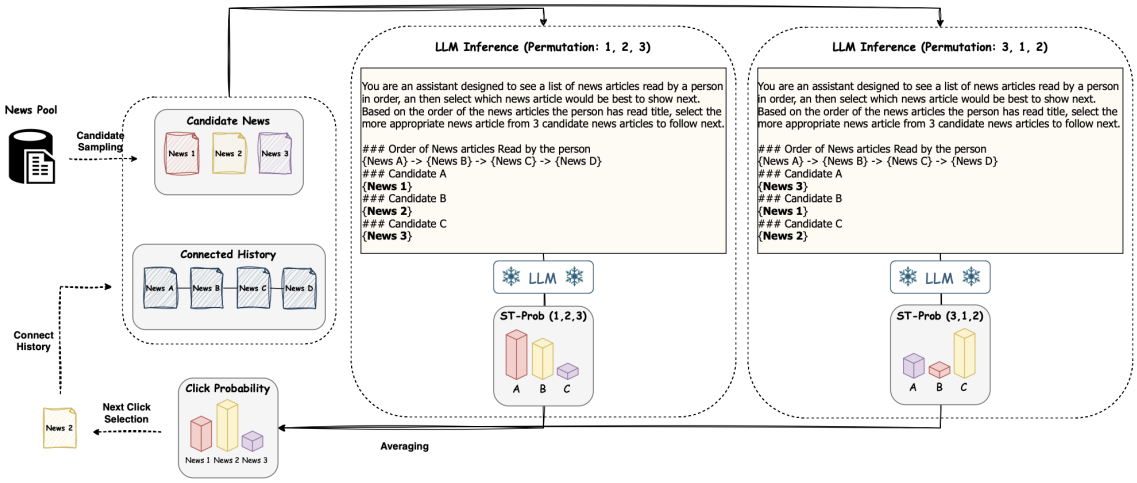


Figure 1: The overall process of user simulation. This figure illustrates an example at the 4th time step with 3 candidates.

2 Related Works

News Recommendation. News recommendation presents unique challenges distinct from other domains due to the highly time-sensitive nature of news content; as new events unfold, articles rapidly become outdated, and users' interests evolve swiftly [27]. This dynamic environment renders traditional approaches, such as collaborative filtering based on static IDs, less effective. As a result, content-based methods have become prominent in news recommendation systems. These approaches generate news embeddings from article content and user embeddings from click histories, and recommend relevant news by comparing the similarities between them [2, 15, 21, 32, 33]. This approach captures user preferences and the semantic structure of news while enabling efficient real-time inference. In our work, we utilize NRMS with DistilBERT as the news recommender [23, 34, 36].

Zero-Shot Recommendation. Obtaining large-scale user interaction data remains challenging, motivating research into zero-shot recommendation methods. One major line of research focuses on domain transfer, training models on a source domain where real data is available and transferring knowledge to a target domain with limited or no data [6, 9, 10]. Another promising direction leverages LLMs for recommendation tasks [20]. LLMs can understand and generate human-like text, making them valuable for generating recommendations based on content without prior user interaction data. In this work, we extend this idea by using LLMs to simulate user interactions for training a lightweight news recommender.

3 Method

Our proposed framework consists of two main stages: (1) simulating user click data using an LLM, and (2) training a news recommender.

3.1 User Simulation using LLM

In the user simulation stage, we use an LLM to simulate user clicks and generate candidate news articles along with their click probabilities. Given a pool of news articles $\mathcal{N}$, we aim to simulate a user click history $N^u = [n_1^u, n_2^u, \ldots, n_M^u]$ over M time steps. At each time step i, we sample a set of L candidate news articles $N_i^c = [n_{i,1}^c, n_{i,2}^c, \ldots, n_{i,L}^c]$ from the pool $\mathcal{N}$. Using the LLM, we estimate the click probability vector $P_i = [p_{i,1}, p_{i,2}, \ldots, p_{i,L}]$, where $p_{i,j}$ represents the likelihood that the user clicks on candidate $n_{i,j}^c$ given their history $N_i^u = [n_1^u, n_2^u, \ldots, n_i^u]$. Based on the estimated probabilities P_i, we select the next news article n_{i+1}^u to connect the user history, simulating the user's click behavior at step $i+1$. This iterative process continues until we construct a complete simulated user history of length M. Figure 1 illustrates the overall process of user simulation. The prompt we used in the experiment can also be found in Figure 1.

Candidate Sampling. At each step of the simulation process, we employ two strategies for candidate sampling to generate the set of candidate news articles N_i^c. The first strategy is random candidate sampling, where we randomly select L news articles from the news pool $\mathcal{N}$. This approach ensures a diverse set of candidate articles by avoiding bias toward any specific topic or theme. The second strategy is embedding-based candidate retrieval. For each news article n_j^u in the history, we obtain its embedding h_j^u using a text embedding model; specifically, we use the Contriever [12] model in this work. We then aggregate these embeddings using mean-pooling to create a single user embedding h_i^u. Using h_i^u, we retrieve L candidate news articles from the news pool $\mathcal{N}$ that are most similar in the embedding space. This method generates candidate sets that are thematically related to the topics present in the simulated history, making the simulation more focused on the specific content trends observed in the generated user behavior.

LLM Inference & Next Click Selection. At each time step, we use the LLM to estimate the user's likelihood of clicking each candidate news article based on their simulated history. To reduce biases arising from the order in which candidates are presented to the LLM, we randomize the candidate sequence and perform inference T times. For each permutation $t \in 1, 2, \ldots, T$, we obtain probability estimates $P_i^{(t)}$. The final click probability for each candidate is

Table 1: Statistics of the benchmark datasets.

	MIND			Adressa			EB-NeRD		
	Train	Test	LAUS	Train	Test	LAUS	Train	Test	LAUS
# Impression	157k	73k	500k	294k	253k	500k	233k	245k	500k
# User	50k	50k	20k	114k	73k	20k	15k	15k	20k
# News	51k	42k	68k	20k	20k	57k	12k	21k	105k

computed by averaging these estimates:

$$p_{i,j} = \frac{1}{T} \sum_{t=1}^{T} p_{i,j}^{(t)}$$

where $p_{i,j}^{(t)}$ is the probability assigned to candidate $n_{i,j}^{c}$ in the t-th permutation. In each LLM inference, inspired by the FIRST method [22], we employ a single-token probability decoding technique to efficiently estimate these probabilities. Candidate probabilities are computed by extracting the output probabilities of candidate identifier tokens during the generation of the first token. This approach effectively approximates the full candidate probability distribution with minimal computational overhead. With the final probability vector P_i, we simulate the user's next click n_{i+1}^{u} by sampling from this distribution. The selected article is then appended to the user history, forming the updated history for the next time step. This process iterates, building a simulated user history that reflects the LLM's understanding of user behavior patterns.

3.2 Training News Recommender

In this stage, the goal is to train a lightweight news recommender using the simulated data generated from the previous stage. Unlike other news recommender systems that rely on click logs with binary indicators (clicked or not clicked), our approach leverages the full probability distributions over candidates at each time step, as provided by the LLM. This rich probabilistic information enables us to train the model using the KL-div loss, allowing for a more nuanced learning process. At each time step i, the model utilizes the user history N_i^u and the candidate set N_i^c to predict the click probabilities for the candidates, outputting a probability distribution $\hat{P}_i = [\hat{p}_{i,1}, \hat{p}_{i,2}, \ldots, \hat{p}_{i,L}]$, where $\hat{p}_{i,j}$ is the predicted probability that the user will click on candidate $n_{i,j}^{c}$. Given the target probability distribution $P_i = [p_{i,1}, p_{i,2}, \ldots, p_{i,L}]$ obtained from the LLM inference, the training objective is to minimize the KL divergence between the target distribution P_i and the model's predicted distribution $\hat{P}i$:

$$\mathrm{KL}(P_i \parallel \hat{P}i) = \sum_{j=1}^{L} p_{i,j} \log \left(\frac{p_{i,j}}{\hat{p}_{i,j}} \right)$$

By minimizing this divergence, the model learns to produce predictions that closely align with the LLM-generated probabilities, effectively capturing the underlying recommendation patterns.

4 Experiments

4.1 Experimental Setup

Datasets. We evaluated our proposed framework using three news recommendation datasets: MIND [37], Adressa [8], and EB-NeRD[14], which contain news articles in English, Norwegian, and Danish, respectively. MIND is collected from the Microsoft News website; we used the MINDsmall-dev set for evaluation and MINDsmall-train set for training supervised models. For the Adressa dataset, sourced from Adresseavisen, we utilized the Adressa-oneweek data and processed the train and test sets following the preprocessing techniques outlined in Yi et al. [39]. EB-NERD, gathered from Ekstra Bladet, includes Danish news articles; we employed EB-NERD-small-validation for evaluation and EB-NERD-small-train for supervised model training. Key statistics of these datasets are summarized in Table 1.

Implementation Details. We simulated 10,000 users in the user simulation phase using different candidate sampling methods; for each user, at every time step, we set the candidate length $L = 20$ and performed $T = 3$ permutations to estimate click probabilities, averaging the results to mitigate order bias. The click history length was capped at $M = 25$. For the simulation, we utilized news articles from MINDlarge for English, Adressa-threemonth for Norwegian, and EB-NERD-large for Danish, ensuring complete separation from evaluation data by excluding any articles present in the evaluation datasets. We employed the Llama-3.1-70B model as the LLM for simulation. Training was conducted with a batch size of 64 and a learning rate of 5×10^{-5}, using the AdamW optimizer; we applied early stopping based on validation loss, splitting a portion of the training data for validation and halting training after three epochs without improvement.

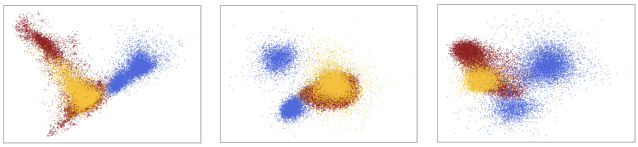
Baselines. In our comparative experiments, we evaluated several baseline methods alongside our proposed framework to benchmark its performance. **BM25** computes BM25 scores between the news articles in the user's history and the candidate articles, recommending candidates based on these relevance scores. **Contriever-sim**, utilizes Contriever [12] to embed each news article. For user embedding, it aggregates embeddings in the user's history via averaging. **UnisRec** [10], leverages item description texts to learn transferable representations across different recommendation scenarios. **LLaMA** [7] re-ranks candidates by inputting the user's history and candidates into the LLaMA model in a manner similar to RankGPT [25], and **LLaMA-ST** employs the prompt illustrated in Figure 1 to input the history and candidates, ranking them based on the probabilities of the first single token. Specifically, LLaMA-8B and LLaMA-70B are based on the LLaMA-3.1 series, whereas LLaMA-1B is based on the LLaMA-3.2. To simulate a scenario where target language data (e.g., English MIND dataset) is absent, we implemented **NRMS-translated**, which entails translating datasets from other languages (such as Adressa in Norwegian and EB-NERD in Danish) into the target language using GPT-4o, and then training the news recommender on this translated data.

4.2 Experimental Results

Main Results. Table 2 summarizes the results, highlighting the MRR, NDCG@5, NDCG@10, and latency. (1) Our proposed framework, LAUS, demonstrates significantly higher performance compared to zero-shot recommendation methods, while maintaining similarly low latency. (2) Compared to LLaMA-based zero-shot recommendation approaches, LAUS offers a substantial reduction in latency without compromising on performance. While LLaMA-70B-ST achieves competitive results, it does so with a latency of 816.68 ms/rec, which is significantly higher than LAUS. (3) LAUS

Table 2: Overall performance comparison of our proposed LAUS framework with baseline methods on three datasets. Note that for the MIND dataset, results for LLaMA-ST are not reported because the number of candidates exceeds 26—the total number of alphabet letters—making it incompatible with LLaMA-ST's use of identifiers and thus precluding a fair comparison.

Model	MIND			Adressa			EB-NeRD			Latency
	MRR	NDCG@5	NDCG@10	MRR	NDCG@5	NDCG@10	MRR	NDCG@5	NDCG@10	(ms/rec)
Zero-Shot Recommendation										
BM25	0.2594	0.2766	0.3377	0.244	0.2317	0.3057	0.3243	0.3585	0.4416	2.85
Embedding-sim	0.2876	0.3124	0.3744	0.2878	0.2798	0.3425	0.3306	0.3665	0.4474	5.19
UniSRec	0.2219	0.2285	0.2916	0.1841	0.1524	0.2328	0.3151	0.348	0.4323	12.96
LLaMA-based Zero-Shot Recommendation										
LLaMA-3.2-1B	0.2183	0.2231	0.2864	0.1747	0.1421	0.218	0.3127	0.3457	0.4298	447.32
LLaMA-3.2-1B-ST	-	-	-	0.1762	0.1438	0.2194	0.3265	0.3643	0.4518	30
LLaMA-3.1-8B	0.2211	0.2262	0.2883	0.1861	0.1559	0.2296	0.3149	0.3473	0.4316	686.68
LLaMA-3.1-8B-ST	-	-	-	0.2651	0.2519	0.3261	0.3435	0.3834	0.4675	115.08
LLaMA-3.1-70B	0.2826	0.3062	0.3648	0.3042	0.3015	0.3679	0.3205	0.353	0.4377	3554.93
LLaMA-3.1-70B-ST	-	-	-	0.3343	0.3331	0.4005	**0.3501**	**0.3898**	**0.4732**	816.68
Trained NRMS based Recommendation w/o Real Data										
Translated	0.2467	0.2615	0.3221	0.2161	0.1845	0.2707	0.3139	0.3456	0.4308	5.69
LAUS	**0.314**	**0.3442**	**0.4023**	**0.3611**	**0.3715**	**0.4431**	0.3456	0.3816	0.4598	5.69
- *RCS*	0.3066	0.337	0.3958	0.356	0.3703	0.4414	0.3447	0.3807	0.4587	5.69
- *Permutations*	0.3082	0.3356	0.3947	0.3569	0.3696	0.439	0.3444	0.3803	0.4577	5.69
Trained NRMS based Recommendation w/ Real Data										
Supervised	0.3174	0.3451	0.4091	0.3714	0.3647	0.4427	0.3554	0.3986	0.4745	5.69
LAUS + Supervised	**0.3183**	**0.3543**	**0.4169**	**0.3859**	**0.3855**	**0.4468**	**0.3803**	**0.4254**	**0.4966**	5.69



(a) MIND **(b) Adressa** **(c) EB-NeRD**

Figure 2: PCA visualization of user history embeddings. Red dots represent LAUS-simulated data, yellow dots represent real user data, and blue dots represent translated data.

surpasses models trained on translated datasets from other languages (Translated), confirming that our simulated user data is more effective for training than translated real-world data. Even though LAUS is trained solely on simulated data without access to real user interactions, it achieves performance comparable to the supervised model trained on real data. This suggests that our framework can effectively learn user preferences through LLM-simulated interactions.

Pretraining. We explored the impact of combining our LAUS framework with supervised fine-tuning using real user interaction data. First, we pretrained the news recommender on simulated data generated by LAUS from the three datasets, and subsequently fine-tuned the model using the real user interactions from each dataset. The results, presented in the last block of Table 2, demonstrate that combining LAUS with supervised fine-tuning leads to further performance improvements across all datasets. This indicates that pretraining with LLM-simulated data can enhance model performance when real data is available for fine-tuning, presenting a promising direction for leveraging LLMs in news recommendation systems.

Ablation Study. We conducted an ablation study to evaluate the contributions of Retrieval-based Candidate Sampling (RCS) and Probability Permutations in our framework (see Table 2, third block). Compared to the full LAUS, removing RCS—using only random candidate sampling—led to decreased performance across all datasets, indicating that RCS improves candidate quality by retrieving relevant articles and enhancing user simulation. Likewise, eliminating Probability Permutations—relying on a single LLM inference—also reduced performance, showing their importance in mitigating position bias and obtaining reliable click probability estimates. These findings underscore that both RCS and Probability Permutations are essential for effective user behavior simulation in our framework.

Analysis. We evaluated the similarity between LAUS-simulated data and real user interactions by performing Principal Component Analysis (PCA) on embeddings of 10,000 randomly sampled user histories from three datasets: LAUS, real user data, and translated data from other languages. Each history was embedded using Contriever, and the embeddings were reduced to two dimensions for visualization. As shown in Figure 2, the resulting plot showed that LAUS-simulated data clustered closely with real data, while translated data was distinctly separated. This indicates that LAUS effectively simulates realistic user interactions that align with real data in the target language, whereas translated data fails to capture the cultural and linguistic nuances of user behavior.

5 Conclusion

In this paper, we proposed **LAUS**, a framework for training lightweight news recommender without relying on real user interaction data. LAUS simulates user behavior using techniques such as retrieval-based candidate sampling and single-token probability blending, generating rich probabilistic data for training. By applying KL-divergence loss on the simulated user history, candidates, and their click likelihoods, we effectively trained a news recommender. Experimental results demonstrated that, despite the absence of real data, the model trained through LAUS achieved high performance while maintaining low latency.

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